ashwa RACING

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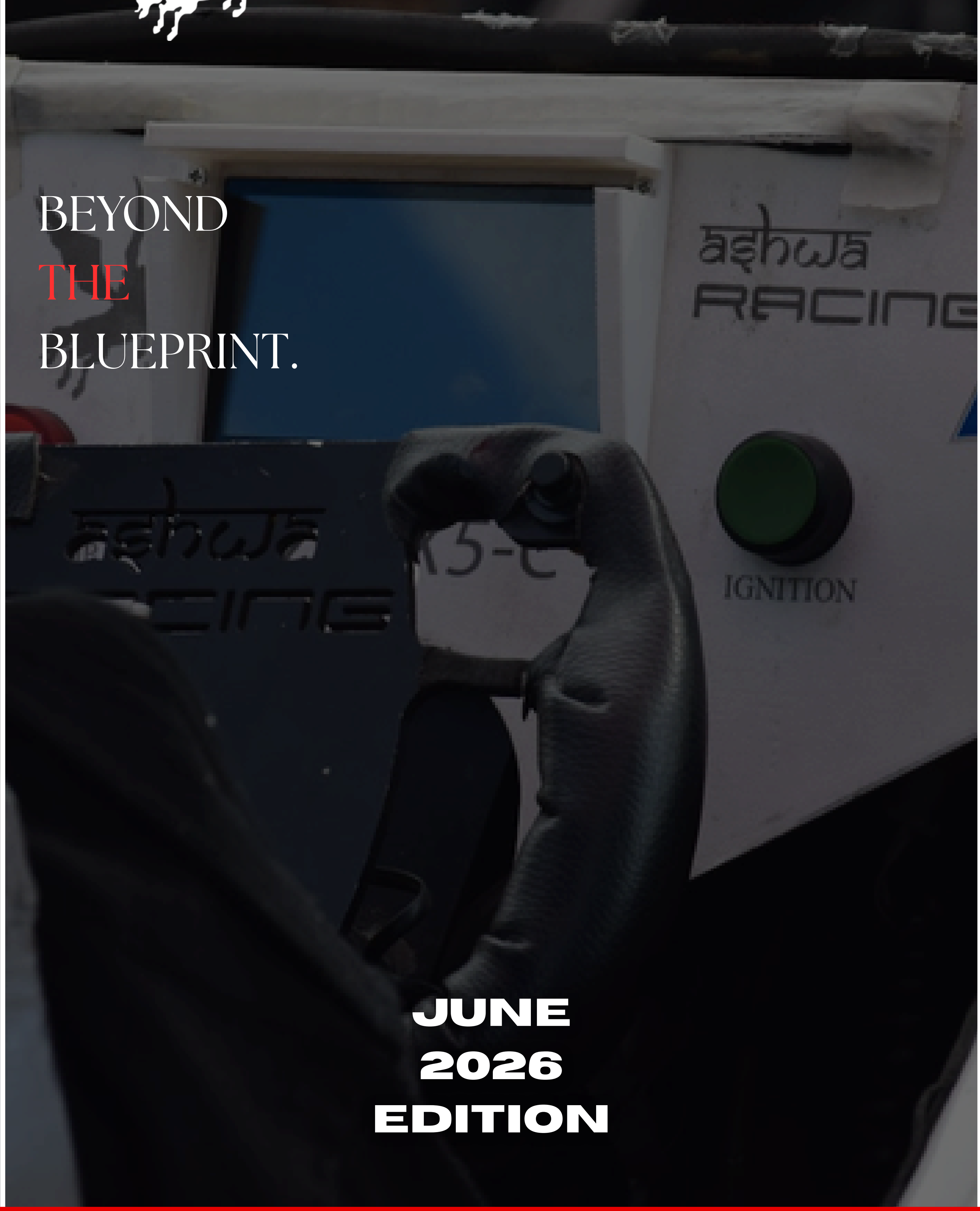
BEYOND THE BLUEPRINT.



JUNE 2026 EDITION

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VISION AND MISSION

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GUIDING PURPOSE,
INSPIRING FUTURE.

MISSION AND VISION

VISION

To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.

To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.

Showcase India’s engineering expertise in mobility on a global stage and provide talent opportunities for the industry.

“ ”

TRACK TALK

THE EVOLUTION OF
THE FORMULA
STUDENT

From a Cancelled Race to a Global Movement: The Evolution of Formula Student

Formula Student's story starts with a competition that almost didn't happen at all. In 1980, Ron Matthews, then an untenured assistant professor at the University of Texas, wanted to enter SAE's Mini-Indy competition with his students, only to learn it was no longer being organised.

Rather than let the idea die, the group decided to start a new asphalt racing competition under a new name, Formula SAE, with Matthews reaching out to SAE International for approval to host it. He and his students ran the competition for its first four years, until he felt it had grown enough to stand on its own. The Society of Automotive Engineers officially began running its Formula SAE programme the following year, in 1981.

The format stayed close to its engineering-education roots, a fictional manufacturing company contracts a student design team to develop a small formula-style race car, which is then evaluated for its potential as a production item, with teams designing, building, and testing a prototype against SAE's rules.

Photo of University of Texas Arlington Formula SAE team with vehicle, 1984.
Source:https://fsae.uta.edu/history/

By the late 1990s, the competition had outgrown its American campus origins. In 1998, three US teams and four UK teams competed in a demonstration event at the MIRA Proving Ground in Warwickshire, with UT Arlington taking the overall win, and the University of Birmingham topping the UK entries.

The Institution of Mechanical Engineers (IMechE) then took on management of this new European venture in partnership with SAE, and the competition has run annually ever since.

For its first two decades, Formula Student was almost entirely an internal-combustion competition, with safety rules evolving alongside the engineering. By the end of its first decade, the wider Formula Student series had already grown to more than 13 countries with over 400 participating teams globally, and performance kept climbing alongside the technology.

Every shift in this timeline followed the same pattern: a small group of students and academics pushed past what the existing rulebook assumed was possible, a cancelled race reborn as a new competition, a demonstration event growing into an annual fixture, a combustion-only grid making room for an electric class and the wider ecosystem caught up around them.

Formula Bharat and the Indian FSAE circuit sit on the far end of this entire timeline, built on decades of incremental pushes by students who came before us. It's a good reminder that the next leap forward usually starts with someone on a team like ours!

Sources:
- History of Formula SAE
- History of Formula Student
- Acknowledging Technology Day 2021 through the Evolution of Formula Student

RZ-XX7C: BEYOND THE ENGINE

EXPLORING THE ELECTRONICS
BEHIND OUR COMBUSTION PROTOTYPE.

RZ-XX7C: BEYOND THE ENGINE

As we move to the end of the RZ-XX6-C post-race testing phase to the manufacturing of the RZ-XX7-C, significant improvements have been incorporated with regard to the design and overall planned execution. The team has learnt a lot and done a lot to rectify the glaring problems with the last prototype.

Coming to the electrical systems, last year's designs presented significant challenges in terms of system integration, packaging, manufacturability, testing, and cost. To address these shortcomings, the team adopted a completely new design philosophy. Rather than developing each subsystem in isolation, the focus shifted toward a system-level approach, with integration, reliability, and ease of manufacturing taking precedence throughout the design process.

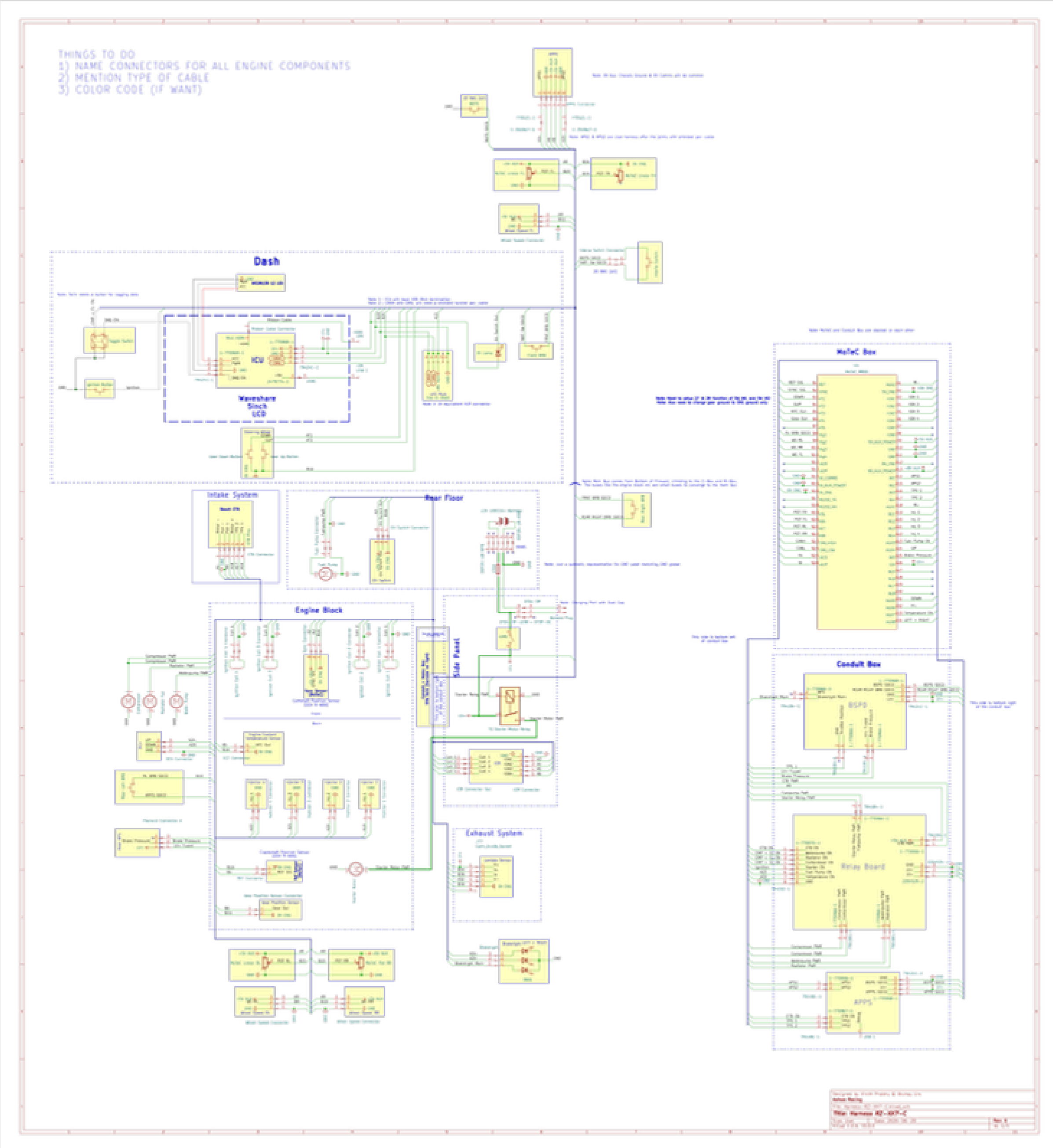
The redesign begins at the heart of our prototype, the MoTeC M800 ECU. In the previous season, our telemetry architecture relied on more than five separate PCBs. The overall design was far from optimal, resulting in lengthy procurement and manufacturing timelines, higher costs, and a final integration that proved difficult to package and maintain.

To address these issues, a clear set of pre-design objectives was established before any hardware development began. This year's architecture was designed to:

- Meet the functional requirements of every subsystem.
- Maximise the utilisation of the features already available within the MoTeC ecosystem.

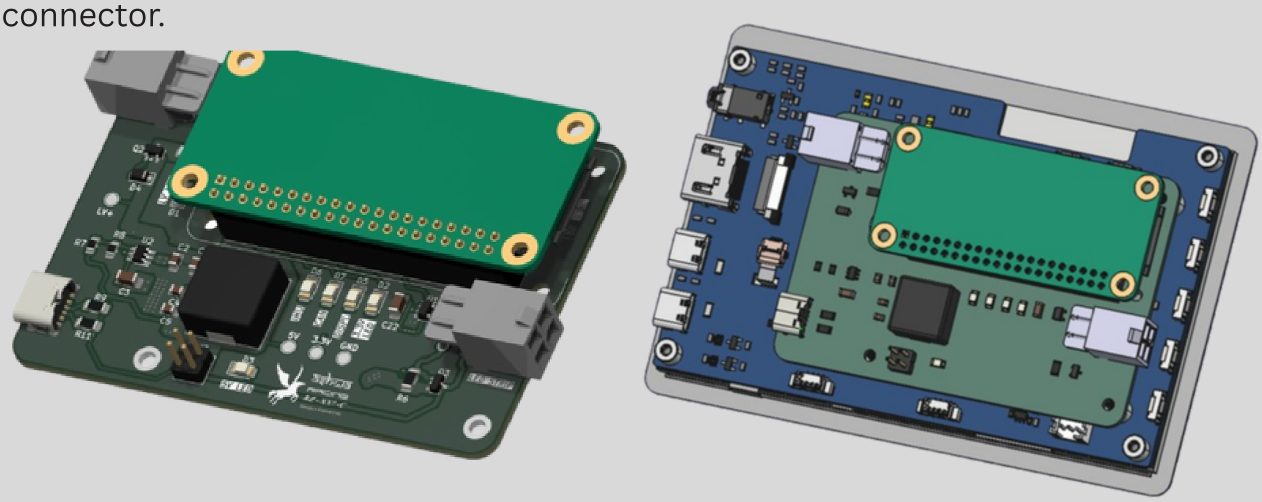
Beyond its conventional functions, the MoTeC M800 is now also responsible for acquiring wheel speed and damper travel data.

All data is transmitted over CAN 2.0 to our Interface Control Unit (ICU), which is our integrated Wireless Telemetry, IMU, Display, and RPM processing unit.



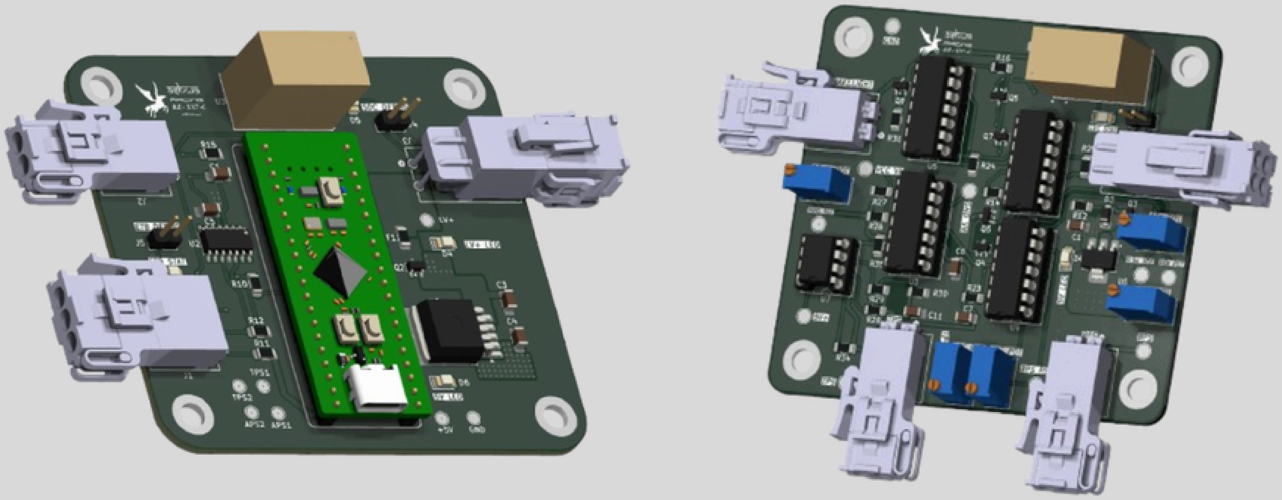
The
Harness Schematic

The Interface Control Unit (ICU) was developed with packaging and integration as the primary design goals. Last year's implementation required three separate boards at the front of the vehicle, adding unnecessary complexity without providing meaningful benefits. The new ICU consolidates these functions into a single compact unit mounted directly behind the display. The ICU is powered directly from the low-voltage (LV+) supply (11–16 V), while the display is powered via the ICU's integrated 5 V buck converter through a USB-C connector.



3D Render of ICU (i) Bare (ii) Behind the display

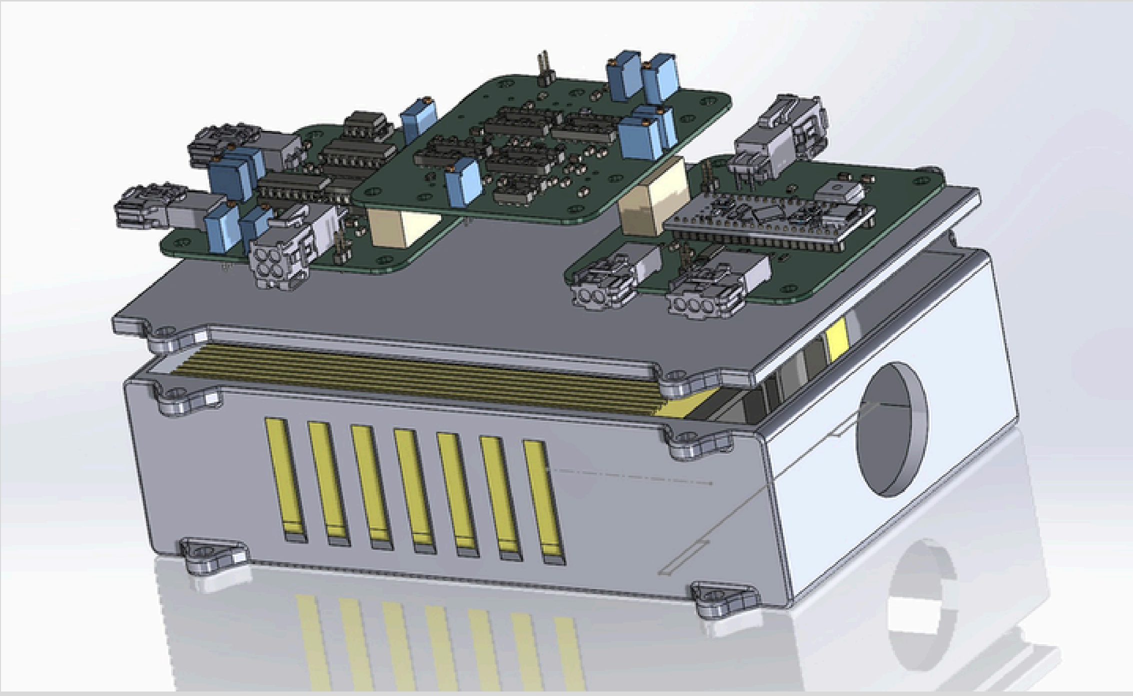
In addition to these changes, the vehicle now utilizes a digital APPS. Compared to the previous analog APPS, the digital one offers improved reliability and a more cost-effective solution.



3D Render of (i) APPS Board (ii) BSPD

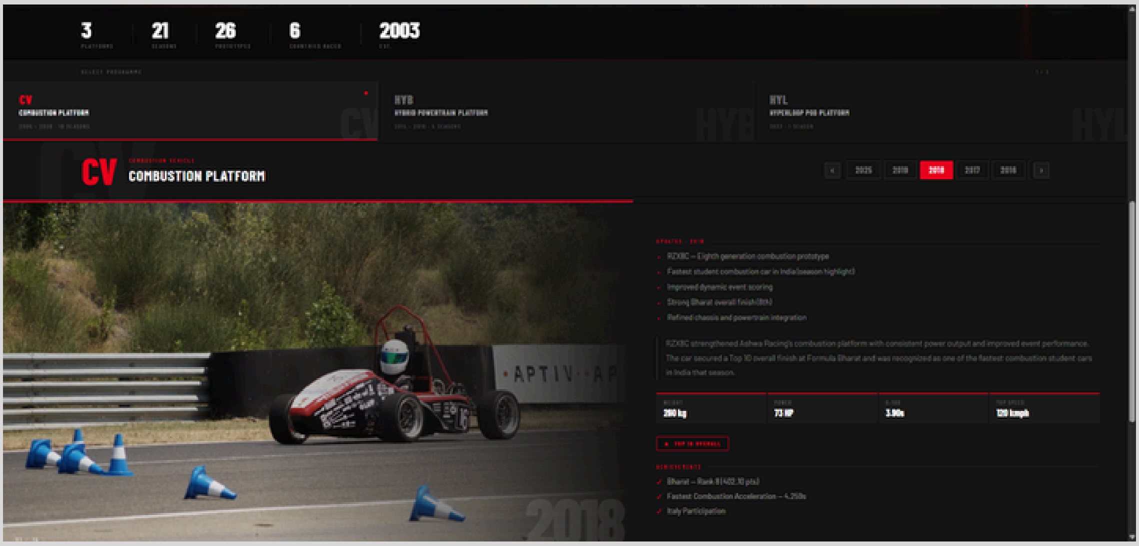
Similarly, BSPD has been designed with integrated brake light control functionality. By incorporating brake light actuation directly into the BSPD, the need for a dedicated brake light actuation board has been eliminated, simplifying the electrical architecture.

We also replaced the power distribution board, which was abominably expensive and only partially functional, with the Relay Board at roughly one-eighth of the cost and twice the functionality. The Relay Board allows us to control all the cooling systems through the MoTeC ECU as well as manually by the driver.



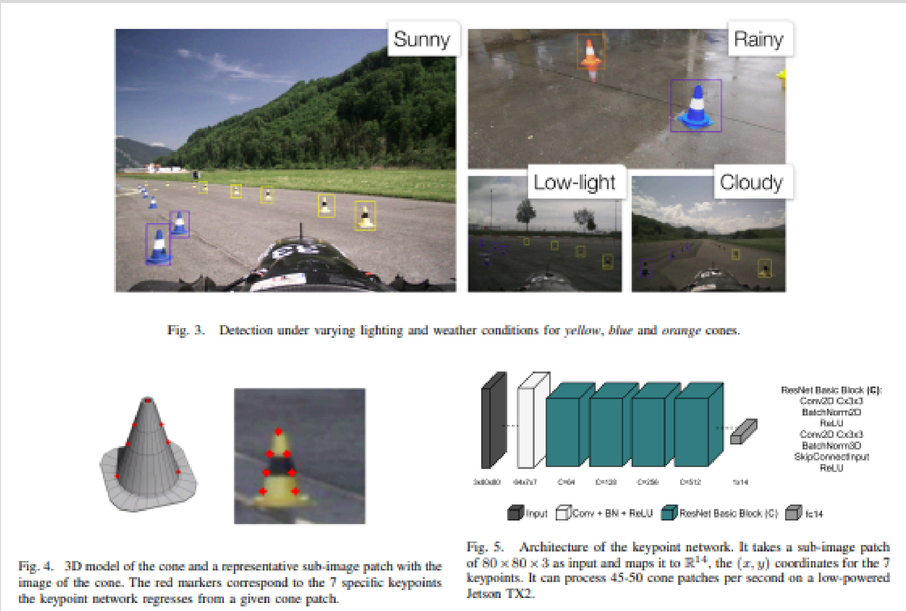
Open-top render of draft MoTeC and Conduit Box

The IT team has been working around the clock to get our website up and running at www.ashwaracing.org The website code is fully open source and available at our organization's GitHub. The website is fully up and logs our current newsletters, along with a gallery with exclusive images, our prototypes from the beginning of the club, and other details. Do visit the website!



Snippet of our website!

Apart from website development, the IT team has been working on a custom ERP software stack while simultaneously advancing the development of our DV systems. Currently, the team is in the pre-design phase, studying research papers, evaluating existing implementations, and analysing their results to establish a strong technical foundation.



Snippet of a paper we are referring

With a rapidly evolving technological landscape, there is always something new to explore. Ashwa Racing provides the perfect platform to experiment with emerging technologies, enabling members to transform theoretical knowledge into practical engineering solutions.

“

RZ-XX8E: VIRTUAL TRACK, REAL RESULTS

”

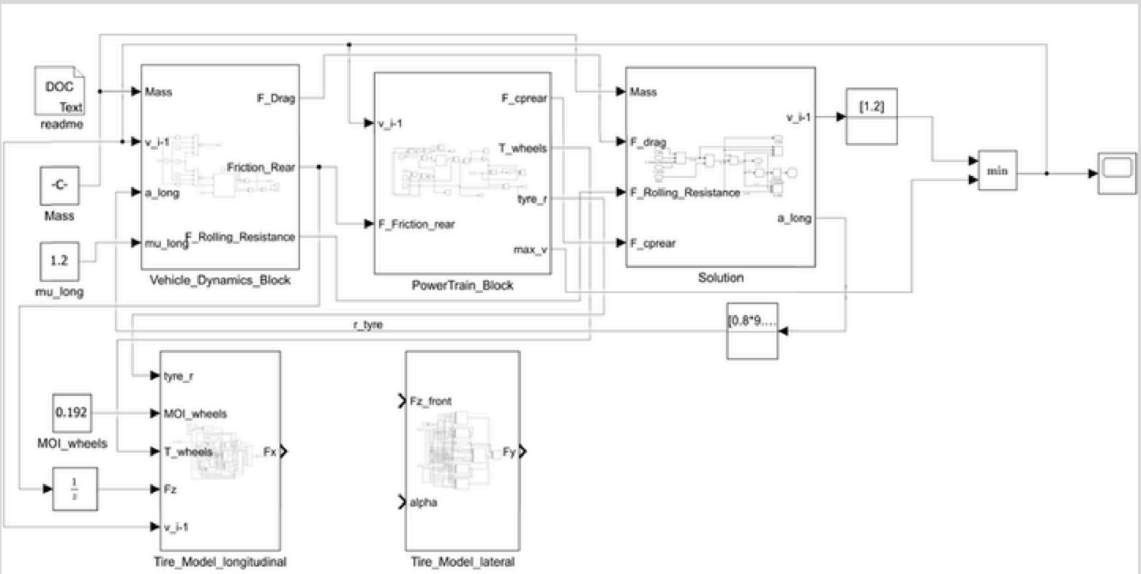
WHERE SIMULATION
MEETS PERFORMANCE

RZ-XX8E: VIRTUAL TRACK, REAL RESULTS

Over the course of June, work centered on various simulations to guide energy calculations across subsystems and setting competition goals for the EV prototype. Members from various sub-systems took up the task for simulations, keeping the team updated regularly on progress.

The first step was competition goal-setting, benchmarked against 2025 FSAE-A results and using a 300kg weight (with driver) as the baseline assumption. Any higher mass would require a honeycomb impact attenuator and not foam, which we already possess. We then moved on to the Energy Estimation, which also decides the Final Drive Ratio(FDR) of our vehicle.

FDR determines the top speed and torque of the race vehicle and how the total energy stored in your accumulator is transmitted to your wheels. It’s an intricate balance; accounting exclusively for only one event (say endurance) would mean compromising on performance in the other events(Autocross, Acceleration and Skid-pad). Hence, the OptimumLap application(a point mass assumption simulation model)was used to determine the optimal values that would keep our performance competitive across the dynamic events. As the name suggests, point mass doesn’t account for any load transfer and consequently any vehicle dynamics. Three iterations were carried out using this preliminary, simplistic assumption.

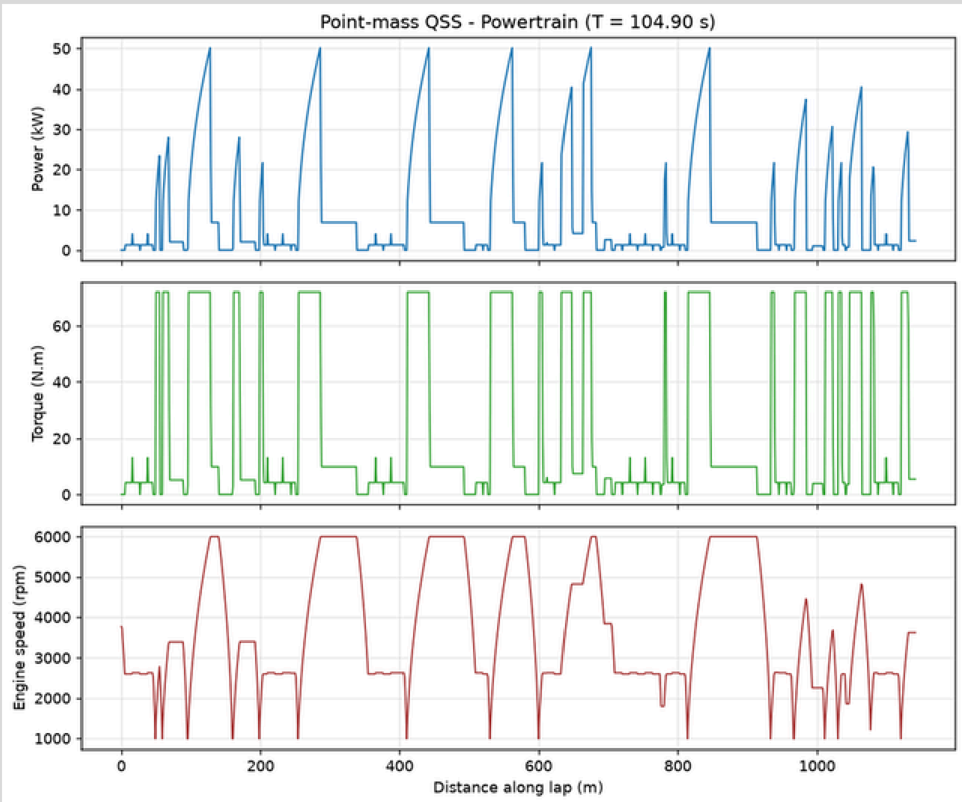


The team's in-house bicycle acceleration model integrating vehicle dynamics, powertrain, and tyre behaviour.

Initial findings highlighted a key constraint: Optimum Lap initially assumed maximum motor torque was available continuously, irrespective of the vehicle's instantaneous operating conditions. As a result, the model predicted significantly higher energy consumption, indicating that an (unrealistically) large accumulator would be required to complete the endurance event. This behaviour was identified as an inherent limitation of the Optimum Lap model, which does not account for throttle variation and therefore overestimates energy demand.

To correct for this, a custom point-mass simulator accounting for throttle variation was used while also accounting for the torque from the speed of the wheels at that instant. Two separate iterations were run. In addition to this, a custom-developed bicycle model (accounting for longitudinal weight transfer) in Simulink was also utilised.

With the throttle variation incorporated, energy requirements dropped significantly. Which essentially means that we can have higher torque caps and comfortably finishing endurance. The motor can output a **continuous torque** of **80 Nm**, with a maximum burst of **110 Nm**. Since the motor is capable of delivering a peak torque, the endurance event was simulated around the continuous torque rating with Autocross, Skid-pad, and Acceleration, utilising this output to maximise performance



Track traces from simulations

With a total car weight of 300 kg, we obtained the following finishing times: **4.6–4.9s** for **Acceleration** (against a 3.9s benchmark, yielding minimum points) and **5.07s** for **Skid Pad** (cross-checked against a 4.995s point-mass result, against a 4.903s benchmark for ~59/75 points). The ideal FDR range sits in the realm of 7-8.

```
% Lateral Model - Local Function
function Fy = mf61_lateral(Fz, alpha, gamma, Vx, ...
                           Fz0, V0, ...
                           pCy1, pDy1, pDy2, pDy3, ...
                           pEy1, pEy2, pEy3, pEy4, ...
                           pKy1, pKy2, pKy3, ...
                           pHy1, pHy2, pHy3, ...
                           pVy1, pVy2, pVy3, pVy4, ...
                           lambda_Cy, lambda_muy, lambda_Ey, ...
                           lambda_Kya, lambda_Hy, lambda_Vy, lambda_muV)

    dfz = (Fz - Fz0) / Fz0;

    Cy  = pCy1 * lambda_Cy;

    muy = (pDy1 + pDy2*dfz) * (1 - pDy3*gamma^2) * lambda_muy / ...
          (1 + lambda_muV * Vx/V0);
```

A snippet of the code used

This concludes our simulations, and we are now moving forward with competition and subsystem-wise goal setting, after which we will transition from our Pre-Design phase into our Design phase. The electrical team is actively working on mapping out the cell configuration and packaging; the drivetrain is working on a feasibility study for the obtained FDR range; and SMP is aiding in cell procurement. Chassis, Suspension, Powertrain and Aero, Brakes continue to sharpen their skills and acquire knowledge by working on the XX6-E.

“

MAY EXPENSES

”

INVESTING IN PERFORMANCE

MAY EXPENSES

EXPENDITURE	AMOUNT
3D Printing of prototype dashboard	₹13,511.00
Reimbursements	₹50,293.00
Total Expenditure	**₹63,804.00**

This period, Ashwa Racing's spending totalled ₹63,804. The bulk of it, ₹50,293, went toward miscellaneous reimbursements across the team, covering the small but essential expenses that keep day-to-day work moving. The remaining ₹13,511 was spent on sandblasting and powder coating for the Electric Prototype's chassis, a crucial step in prepping the 3D-printed prototype dashboard for the EV build.

Together, these numbers reflect the steady, behind-the-scenes work that goes into every subsystem, a proof that progress isn't just measured in lap times, but in the details that make the car race-ready.

“

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”

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